ELENI KARAGIANNI

ARCHITECTURAL ENGINEER & COMPUTATIONAL DESIGNER
SPATIAL DATA · MACHINE LEARNING · MULTI-AGENT SYSTEMS

CONTACT

eleni.karagianni.arch@gmail.com
+44 7586 520983
eleni-karagianni.com ↗
linkedin ↗ · github ↗
London · full UK right to work

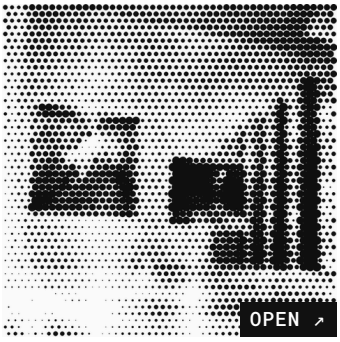
PROFILE

Architectural engineer, licensed in Greece (TEE), with an MArch in Urban Design from the Bartlett. I build computational tools that read and make space: scraping and embedding large multimodal datasets, scoring them with machine learning, and connecting AI agents to live 3D geometry in Blender. I like research where the method is as considered as the result, and where every output can be traced back to the data and decisions behind it.

RECOGNITION

Best Computation Prize, UCL 2026
Theory Report Prize, UCL 2026
Hon. Mention, SCI-Arc, LA 2026
IKY Excellence Award, Greece

SELECTED WORK · click an image to open it



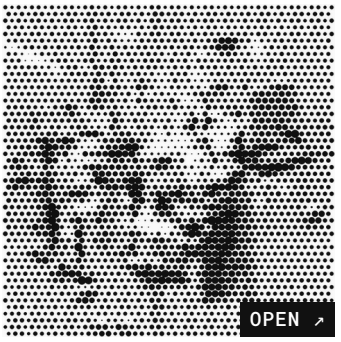
W01 NINE ACTS OF OSTRANENIE ↗
2026 · 9 LLM agents, ML, Blender MCP
with Ruijia Zheng



W02 NIGHT WALK ↗
2026 · street graph, routing
King's Cross to Camden



W03 NINE ACTS LAB ↗
2026 · working demos
archive, agent debate, 3D output



W04 TURIN MACHINE ↗
2025 · LLM agents, TF-IDF, SOM
language-to-form generator

COMPUTATIONAL PROJECTS

2026
The Bartlett, UCL
with Ruijia Zheng
Best Computation Prize · SCI-Arc

NINE ACTS OF OSTRANENIE ↗

- Co-designed and built nine LLM agents on one locally hosted open model (Qwen), each with a deliberately partial view of a 177,241-item archive of Rotterdam literature, photography and film. Led agent design: roles, retrieval, tools, memory and protocols.
- Scraped and unified the archive (38.5k texts, 5.9k photos, 61k film stills); scored it with CLIP embeddings, SOMs and anomaly detection (Isolation Forest + LOF) against a taste model trained on 566 hand-picked items.
- Quantified the language of oddness: descriptions of odd images use 65% more verbs ($p < 0.001$) and 2.5× more negations. The findings set the agents' voices.
- Connected the agents to Blender through MCP in a perception-to-geometry loop: they place cameras in their own models, read the geometry and reshape it. About 100 runs across 14 sites, with a human-in-the-loop curator.
- Built the **web archive ↗**: anomaly-ranked search, semantic paths, map, globe and 3D UMAP views.

2026
Prototype
individual

NIGHT WALK ↗

- Street graph of 8,165 segments from OpenStreetMap and police.uk data, each scored 0–1 on lighting, people around, sightlines, help nearby and incidents. Four walkers route with Dijkstra and agree on one route through minimax-regret negotiation, in the browser.

2025
The Bartlett, UCL
individual

TURIN MACHINE ↗

- Two LLM agents with scripted personas walk 23 locations in Turin and talk about what they encounter.
- Mined the dialogue with TF-IDF, PCA and Self-Organising Maps into a spatial vocabulary. Each word maps to a 3D component, so editing one word regenerates the structure. Also image classification, audio segmentation and 3D printing.

RESEARCH

2026
MArch dissertation
Theory Report Prize

THE UNSEEN CITY ↗

- How dataset composition, thresholds and model choice shape what AI systems "see" in cities and what they leave out. I call this the politics of noise: what falls below a system's threshold gets designed around as if it were absent.
- Tested on MIT's Streetscore and on Nine Acts, with formal agreement and divergence scores between human and machine judgement. Proposes computational systems that declare their partial point of view instead of claiming neutrality.

EXPERIENCE

Oct 2026
London

WORKSHOP CO-LEAD, THE BARTLETT, UCL

"Designing with Disagreement", B-Pro two-week workshop, with Ruijia Zheng

- Wrote the course from scratch, including the Jupyter notebook templates that let 23 postgraduate students, most with no coding background, build working multi-agent systems in two weeks.
- Teams scraped and curated their own datasets, built agents grounded only in that material, tuned personas and thresholds, and ran opposing agents across text, image and 3D through a Blender MCP connection.

2023–25
London

URBAN SAFETY RESEARCH & ADVISORY, LOVME.AI

- Early research behind the "Safety Fit Score" for route safety across the UK; planned and ran user interviews; advised on how street design, lighting and activity shape perceived and actual safety.

2023
Thessaloniki

ARCHITECTURAL INTERN, URBAN SOUL PROJECT

- Produced drawing sets for 5+ projects in parallel (surveys, 3D models, architectural, electrical and plumbing plans), including workplace interiors for PwC's Athens campus.

2023–24
Thessaloniki

ARCHITECTURAL DESIGNER, ANGELOS KOUKOULIS

- Adaptive reuse of commercial buildings into housing: surveys, drawings, visualisation, coordination with engineers and clients.

TECHNICAL SKILLS

PYTHON & DATA	pandas, scikit-learn, SQLite, Jupyter; scraping with requests, BeautifulSoup, Playwright and APIs; cleaning and structuring multimodal datasets (100k+ items)
MACHINE LEARNING	CLIP embeddings, Isolation Forest, LOF, Self-Organising Maps (MiniSom), UMAP, PCA, TF-IDF, logistic regression, significance testing; image classification, audio segmentation
LLMS & AGENTS	Multi-agent orchestration, tool use, memory, retrieval design, human-in-the-loop; local models (Qwen, LM Studio), OpenAI and Claude APIs; Model Context Protocol (MCP)
SPATIAL & GRAPHS	OpenStreetMap and open crime data, street-network scoring, Dijkstra routing, multi-criteria negotiation; site analysis and fieldwork
GEOMETRY & 3D	Blender + Python API (bpy), Rhino, Revit, AutoCAD, 3ds Max, SketchUp, 3D printing
WEB & VISUALISATION	HTML, CSS, JavaScript, Plotly, interactive maps and 3D data views, TouchDesigner; Adobe CC, DaVinci Resolve, Enscape, D5, Twinmotion

EDUCATION

MArch Urban Design
The Bartlett School of Architecture, UCL · 2025–26

Integrated Master's in Architectural Engineering
University of Thessaly · 2020–25 · 9.3/10, top of class
Thesis: Under the Night Veil, the night city through the female lens (9.82/10)

Licensed Architect-Engineer
Technical Chamber of Greece (TEE)

ALSO

Greek (native)
English (proficient)

MITx: Becoming an Entrepreneur, 2021
DelftX: Urban Design for the Public Good, 2024

Nine Acts process blog ↗
Turin Machine process blog ↗